# Midterm 2 for COSC330/PHYS306

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## 1 Instructions

1. Keep it simple: answers involving complex logic circuits may be given using block diagrams. For example, Fig. 1 depicts a 4-bit adder without showing individual XOR gates. If the exercise calls for single gates, however, provide the design at that level of detail.
2. Use WaveDROM whenever possible to create timing diagrams and gate designs.
3. Write your answers in a separate document.
4. This exam is open-book, open-note, to be completed individually.

## 2 Chapter 6 - Functions of Combinational Logic

1. Consider Fig. 1, in which a 4-bit adder is depicted. Assuming a uniform worst-case delay of 8 ns from carry-in to carry-out for each stage, the total delay is 32 ns. (a) At what maximum frequency can this circuit perform additions, if the delay must be less than a clock period? (b) What would the result in (a) be if the adder was extended to 8 bits? (c) Create a timing diagram showing the addition of the numbers in Fig. 1. **Bonus:** include the timing delays, and indicate the clock period.

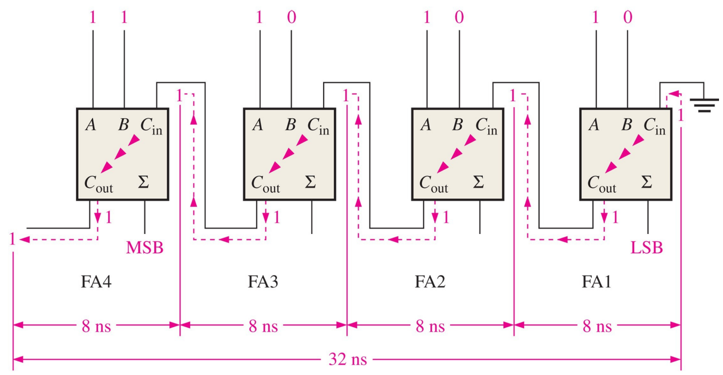



Figure 1: A schematic of a 4-bit adder with delays marked in nanoseconds.

2. Using AND, OR, XNOR, and inverter gates, (a) create a 2-bit comparator that yields True if $A > B$, (b) True if $A = B$, and (c) True if $A < B$. For each circuit, assume both $A$ and $B$ are 2-bit binary positive numbers. (d) Wrap this all into one circuit with three outputs and 4 inputs. *Hint: use Karnaugh maps for parts (a)-(c) to simplify the gates.*
3. Generate the logic gates that convert 4-bit gray code to 4-bit binary code, and show that it works with a timing diagram.
4. Consider Fig. 2, in which a decimal to binary conversion circuit is shown. (a) Add logic to the circuit such that it becomes a hexadecimal to binary converter. (b) Draw a logic symbol for this circuit with the correct number of inputs and outputs. (c) Connect two hex-to-bin converters to a symbolic 2-to-1 multiplexer with the correct number of data select line(s) to form a system that can send two-digit hex numbers over one output line.

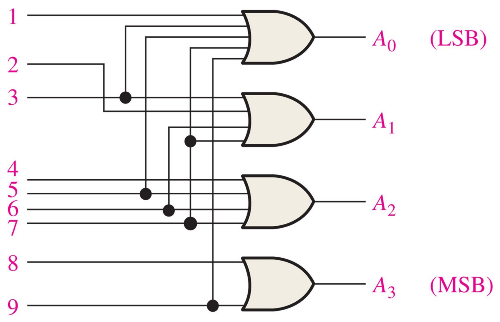



Figure 2: A decimal to binary encoder circuit.

# 3 Chapter 7 - Latches, Flip-flops, and Timers

1. Consider Fig. 3, in which a divide-by-four frequency divider is depicted with D flip-flops and a CLK signal. (a) Elaborate on this circuit to create a circuit that can divide the clock frequency by 2, 4, or 8. Show the flip-flops explicitly. (b) Develop a symbol for the 2-4-8-16 divider, with CLK signal input and four ouputs (one each for $f/2$, $f/4$, $f/8$, and $f/8$, where $f$ is the clock frequency). (c) Connect the symbol for the new part to a symbolic 4-to-1 multiplexer with the appropriate number of data select lines to form a clock division system. (d) Show with a timing diagram the output if the user changes the data select lines at least once. For parts (b)-(d), it is only necessary to show symbols rather than individual flip-flops.

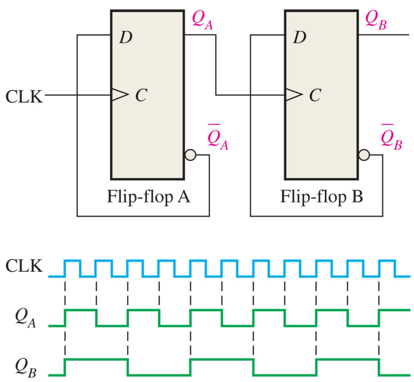



Figure 3: Two D flip-flops forming a frequency divider.

# 4 Chapters 8 and 9 - Shift Registers and Counters

1. Consider the timing diagram in Fig. 4. Using any combination of *shift registers* and supporting gates, create a circuit with 8-outputs that produces this timing diagram.

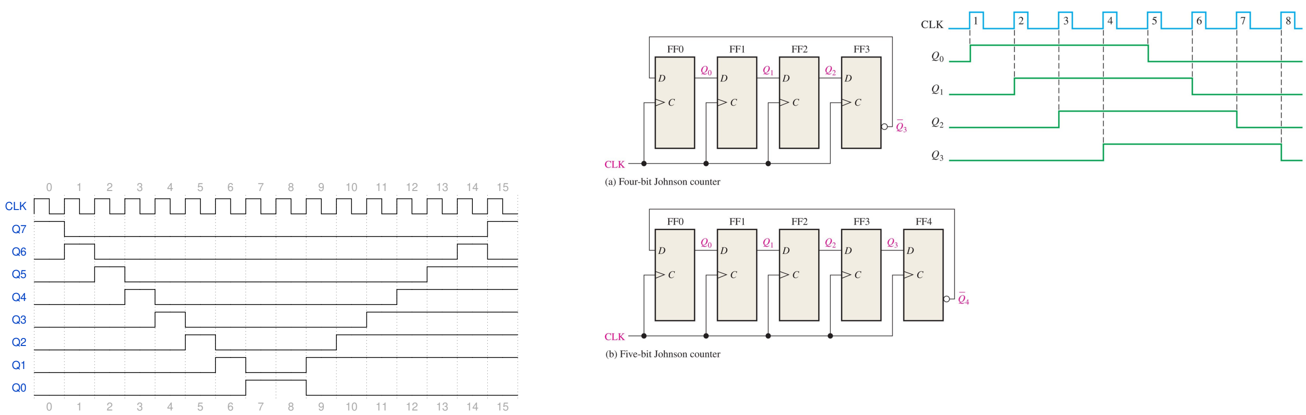



Figure 4: (Left) A waveform that repeats every 16 clock periods. (Right) Johnson counters (4 and 5-bits), and the timing diagram for the four-bit case.

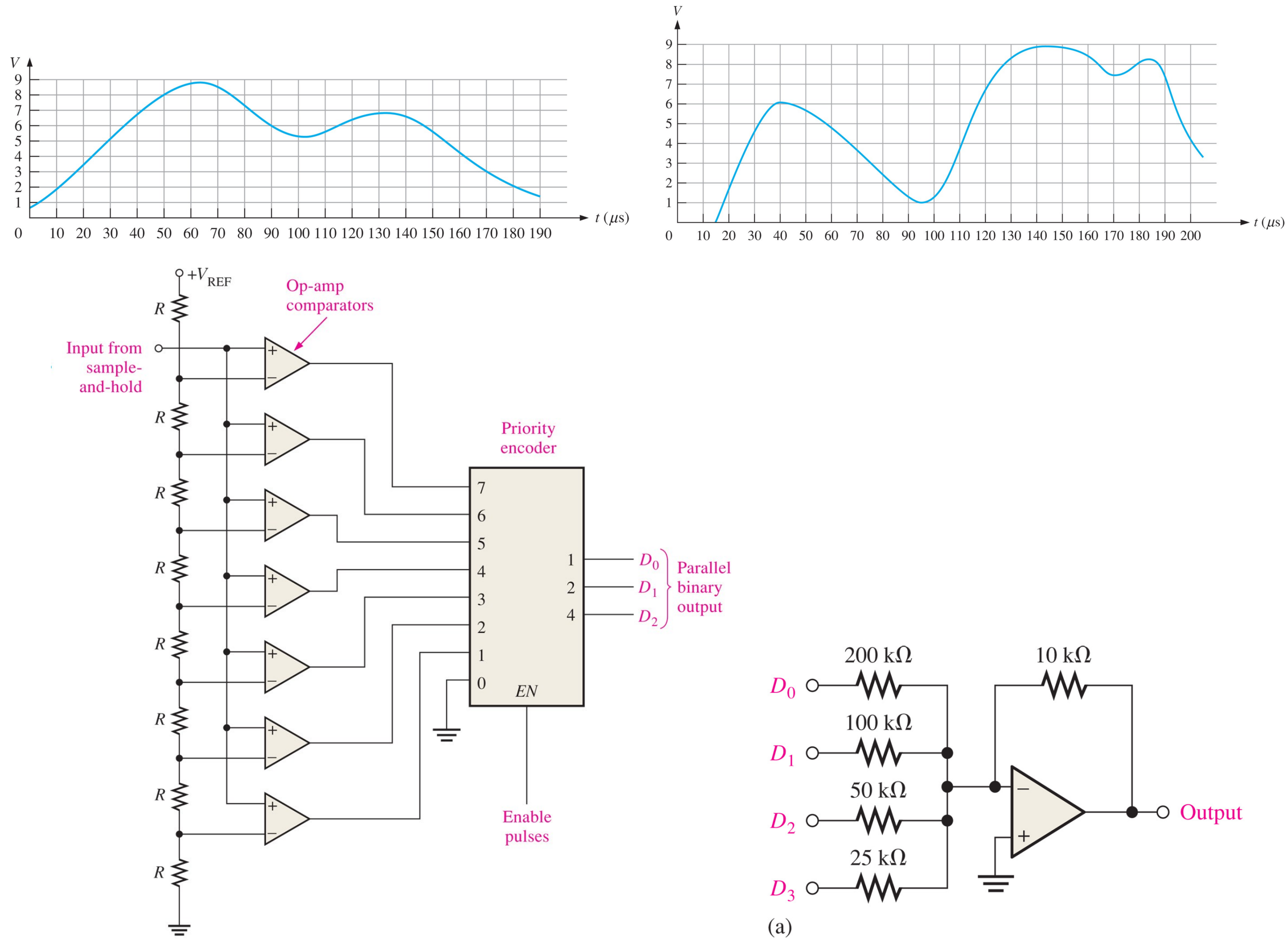



Figure 5: (Top left) An input waveform for a 3-bit flash ADC with $V_{CC} = 8$ Volts. (Top right) A second input waveform for a 3-bit flash ADC with $V_{\text{ref}} = 8$ Volts. (Bottom left) The 3-bit flash ADC. (Bottom right) A 4-bit DAC with binary weighted inputs.

2. Consider Fig. 4 (right), in which 4 and 5-bit Johnson counters are depicted. (a) Create a circuit based on Fig. 4 (right) called a *combinatoric trigger*. Imagine four digital channels A, B, C, and D, as inputs. The output of the circuit should be True if *any two* of the four channels is True *within 100 ns of each other*. Develop any necessary logic symbols, and use the appropriate clock frequency. The Johnson counter(s) can have any number of bits.

# 5 Chapter 12 - Signal Conversion and Processing

1. Consider Fig. 5. (a) Determine the binary output codes from the 3-bit flash ADC in Fig. 5 (bottom left), given the input waveform in Fig. 5 (top left). (b) Repeat (a), but for the input waveform in Fig. 5 (top right).

2. Determine the output of the DAC in Fig. 5 (bottom right), if the sequence of 4-bit numbers is applied to the inputs. The data inputs have a low value of 0V and a high value of 5.0V.

## 2 Chapter 6 - Functions of Combinational Logic

1. Consider Fig. 1, in which a 4-bit adder is depicted. Assuming a uniform worst-case delay of 8 ns from carry-in to carry-out for each stage, the total delay is 32 ns. (a) At what maximum frequency can this circuit perform additions, if the delay must be less than a clock period? (b) What would the result in (a) be if the adder was extended to 8 bits? (c) Create a timing diagram showing the addition of the numbers in Fig. 1. **Bonus:** include the timing delays, and indicate the clock period.

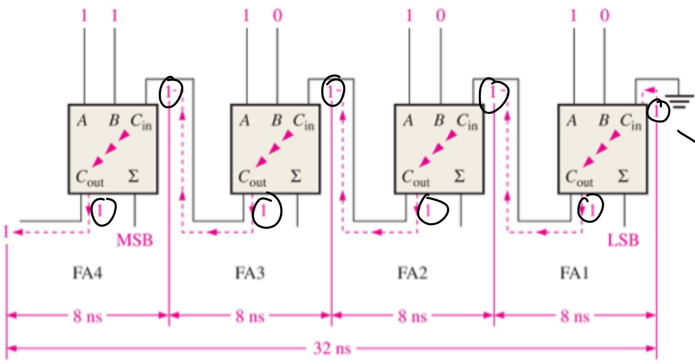



Figure 1: A schematic of a 4-bit adder with delays marked in nanoseconds.

*I'm confused with initial $C_{in}$ value.
Is this question assume $C_{in}$ and $C_{out}$ are High the whole time?

a). since serial connection

total delay time $= 8ns \times 4 = 32ns$

Thus $T_{clk} \geq 32ns$

$$f = \frac{1}{T_{clk}} = \frac{1}{32ns} = \frac{1}{3.2\times10^{-8}s} = 3.125\times10^{7}\,Hz = \underline{\underline{31.25\,MHz}}$$

b). 8 bit serial

$t_{delay} = 8ns \cdot 8 = 64ns.$

$T_{clk} \geq 64ns.$

$$f = \frac{1}{T_{clk}} = \frac{1}{64ns} = \underline{\underline{15.625\,MHz.}}$$

c). $\Sigma = (A \oplus B) \oplus C_{in}$

A: 1111

B: 1000

Initial $C_{in} = 0$

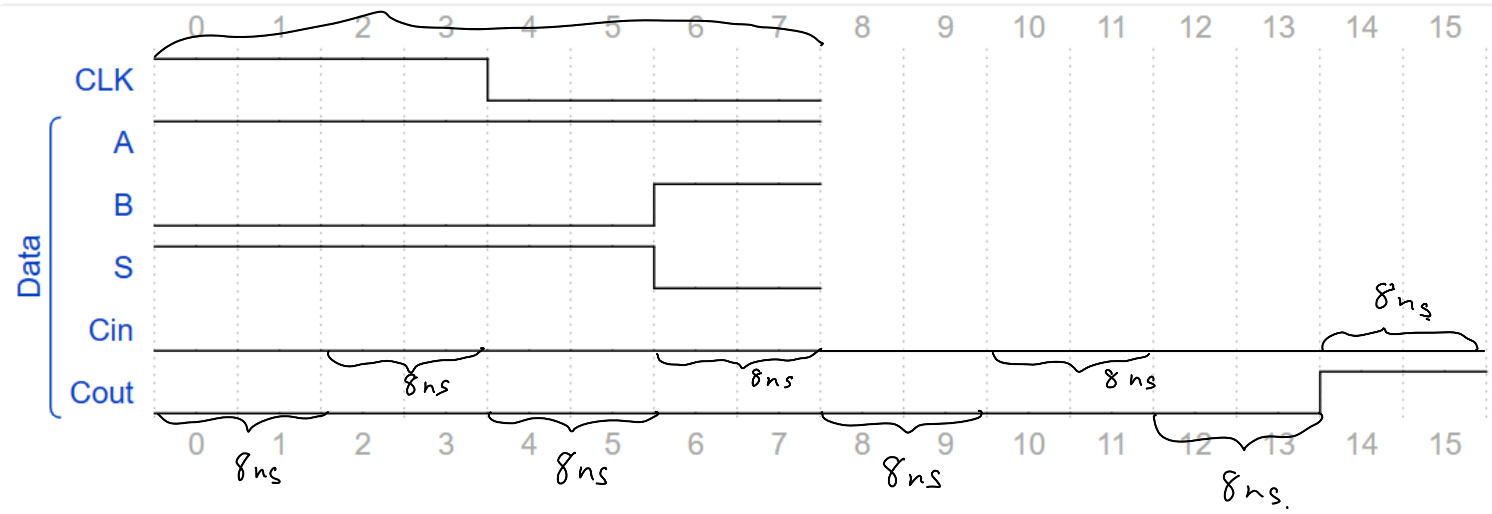
$\Sigma_0 = 1$ $\qquad \Sigma_4 = C_{out} = 1$

$\Sigma_1 = 1$

$\Sigma_2 = 1$

$\Sigma_3 = 0$

32ns Tclock



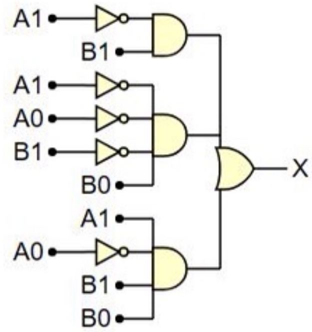

Cin and Cout both have 32ns delay.

2. Using AND, OR, XNOR, and inverter gates, (a) create a 2-bit comparator that yields True if $A > B$, (b) True if $A = B$, and (c) True if $A < B$. For each circuit, assume both $A$ and $B$ are 2-bit binary positive numbers. (d) Wrap this all into one circuit with three outputs and 4 inputs. *Hint: use Karnaugh maps for parts (a)-(c) to simplify the gates.*

a). $A > B$

| A \ B | 00 | 01 | 10 | 11 |
|---|---|---|---|---|
| 00 | | | | |
| 01 | 1 | | | |
| 10 | 1 | 1 | | |
| 11 | 1 | 1 | 1 | |

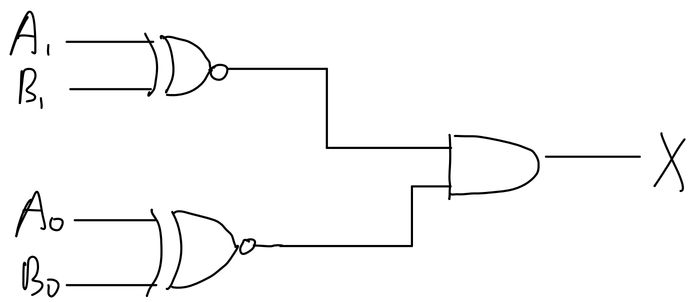
X Active high.




$A_1\overline{B_1} + \overline{A_1}A_0\overline{B_1}\,\overline{B_0} + A_1A_0B_1\overline{B_0}$

b). $A = B$

| A \ B | 00 | 01 | 10 | 11 |
|---|---|---|---|---|
| 00 | 1 | | | |
| 01 | | 1 | | |
| 10 | | | 1 | |
| 11 | | | | 1 |



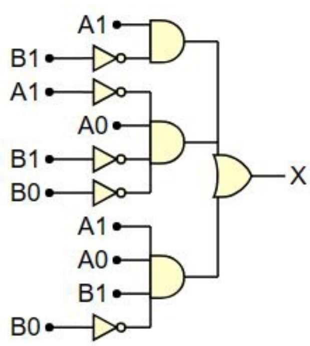

$\overline{A_1}\,\overline{A_0}\,\overline{B_1}\,\overline{B_0} + A_1A_0B_1B_0 + \overline{A_1}A_0\overline{B_1}B_0 + A_1\overline{A_0}B_1\overline{B_0}$

↓

$A_1$ Xnor $B_1$ $\wedge$ $A_0$ Xnor $B_0$

c). $A < B$

| A \ B | 00 | 01 | 10 | 11 |
|---|---|---|---|---|
| 00 | | 1 | 1 | 1 |
| 01 | | | 1 | 1 |
| 10 | | | | 1 |
| 11 | | | | |




$\overline{A_1}B_1 + \overline{A_1}\,\overline{A_0}\,\overline{B_1}B_0 + A_1\overline{A_0}B_1B_0$

3. Generate the logic gates that convert 4-bit gray code to 4-bit binary code, and show that it works with a timing diagram.

$G_3 G_2 G_1 G_0$

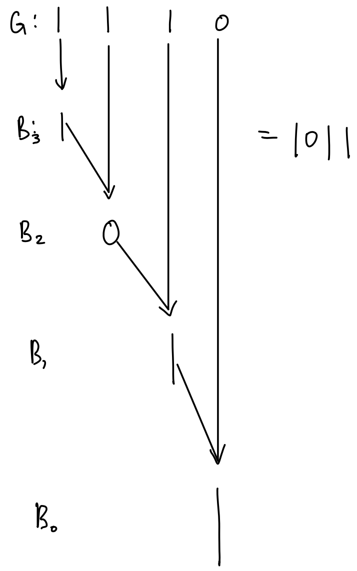
$B_3 B_2 B_1 B_0$

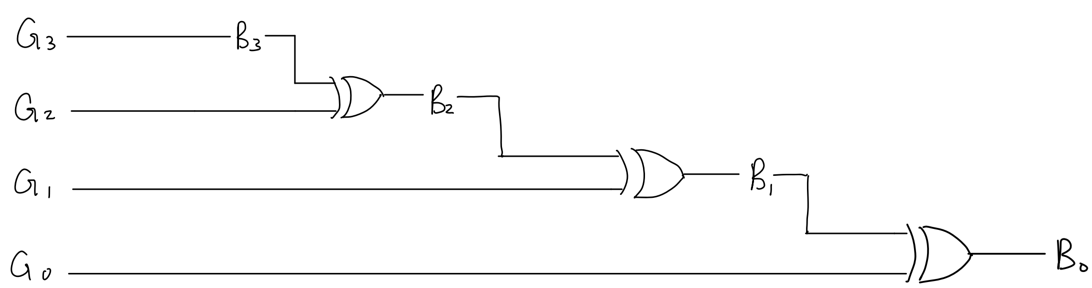
$B_3 = G_3$

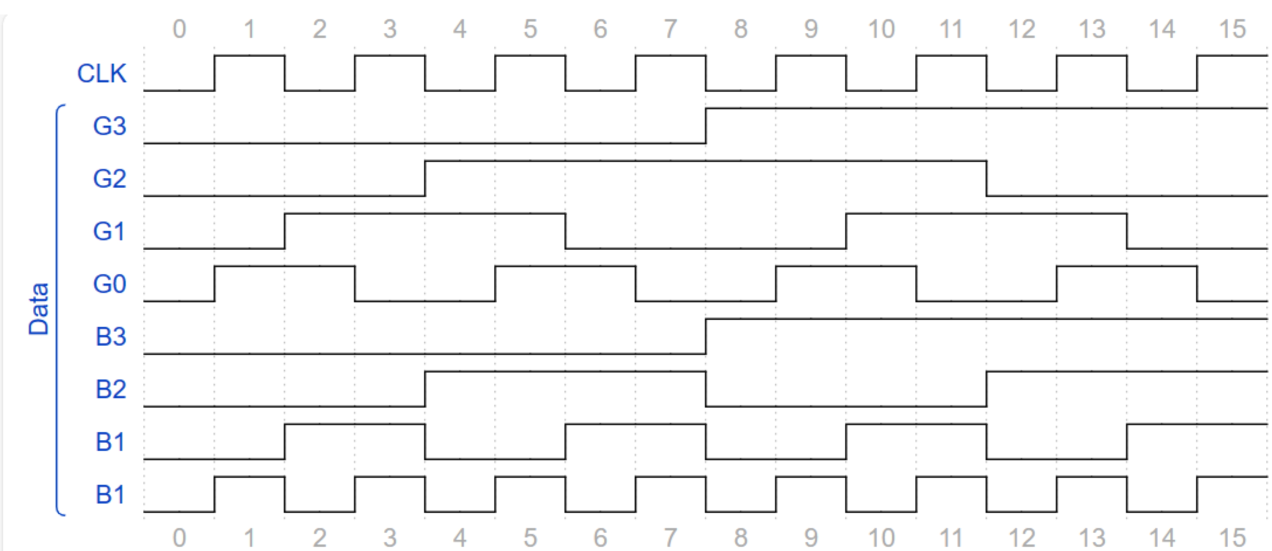
$B_2 = B_3 \oplus G_2$

$B_1 = B_2 \oplus G_1$

$B_0 = B_1 \oplus G_0$



4. Consider Fig. 2, in which a decimal to binary conversion circuit is shown. (a) Add logic to the circuit such that it becomes a hexadecimal to binary converter. (b) Draw a logic symbol for this circuit with the correct number of inputs and outputs. (c) Connect two hex-to-bin converters to a symbolic 2-to-1 multiplexer with the correct number of data select line(s) to form a system that can send two-digit hex numbers over one output line.

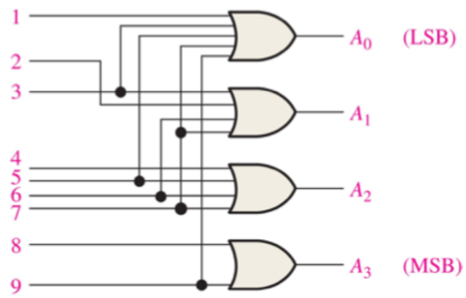



Figure 2: A decimal to binary encoder circuit.

| Hex | Binary ($A_3 A_2 A_1 A_0$) |
| --- | --- |
| 0 | 0000 |
| 1 | 0001 |
| 2 | 0010 |
| 3 | 0011 |
| 4 | 0100 |
| 5 | 0101 |
| 6 | 0110 |
| 7 | 0111 |
| 8 | 1000 |
| 9 | 1001 |
| A | 1010 |
| B | 1011 |
| C | 1100 |
| D | 1101 |
| E | 1110 |
| F | 1111 |

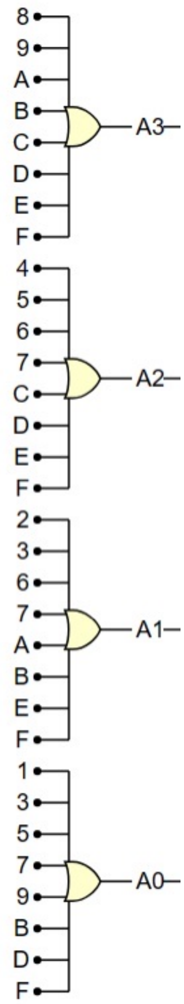
$A_0 = 1+3+5+7+9+B+D+F$

$A_1 = 2+3+6+7+A+B+E+F$

$A_2 = 4+5+6+7+C+D+E+F$

$A_3 = 8+9+A+B+C+D+E+F$

a).




b).

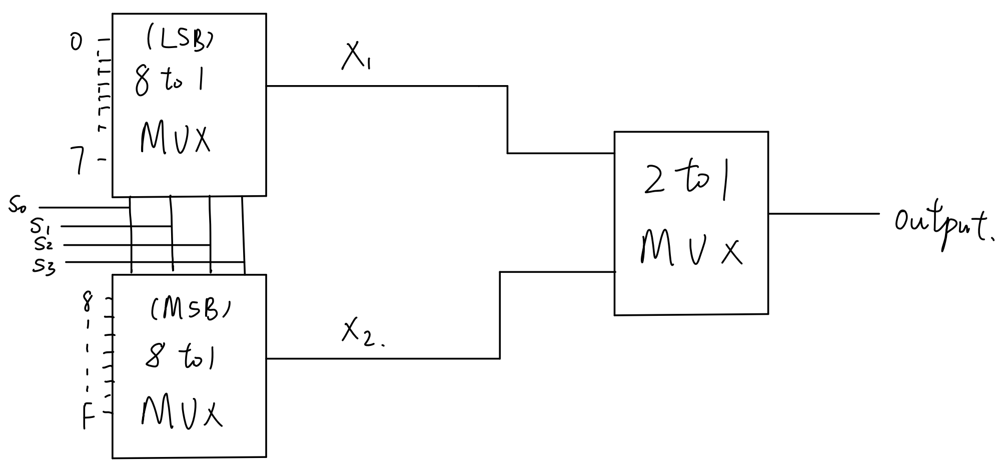
$$A_0 = 1+3+5+7+9+B+D+F$$

$$A_1 = 2+3+6+7+A+B+E+F$$

$$A_2 = 4+5+6+7+C+D+E+F$$

$$A_3 = 8+9+A+B+C+D+E+F$$

C).



## 3 Chapter 7 - Latches, Flip-flops, and Timers

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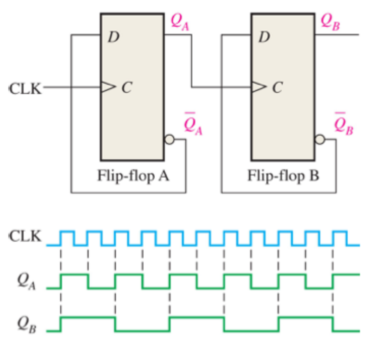
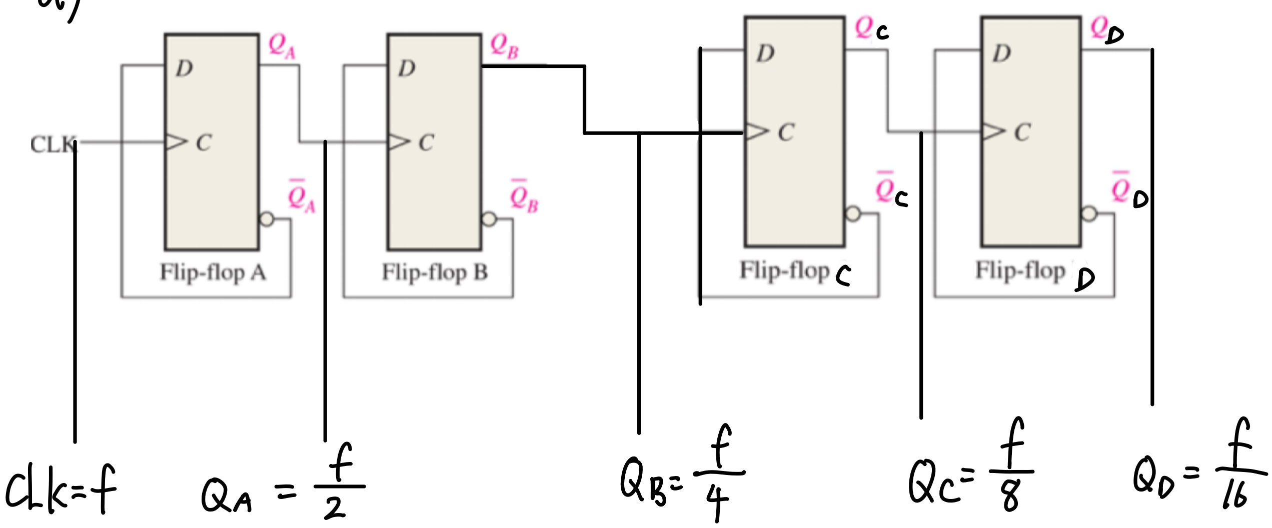
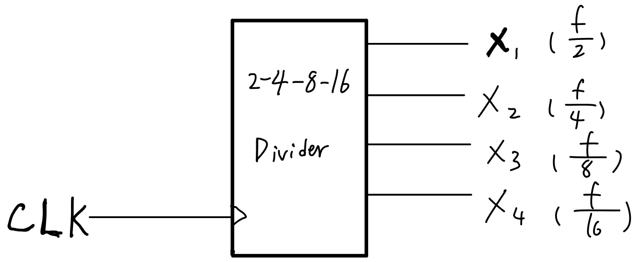



Figure 3: Two D flip-flops forming a frequency divider.

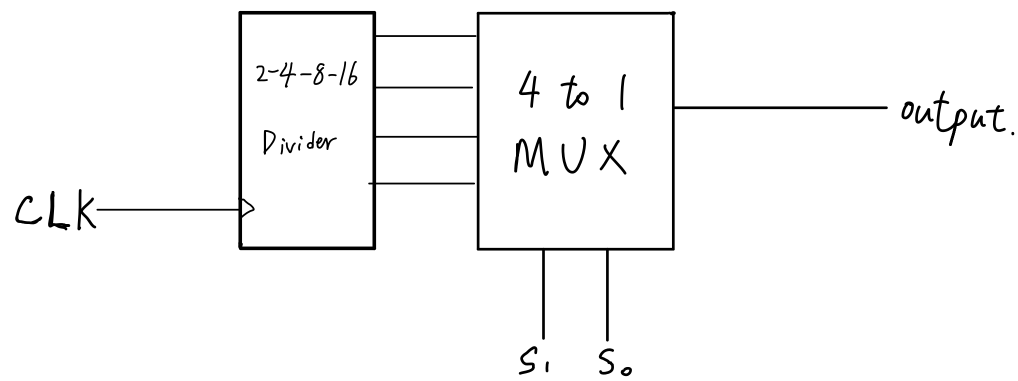
a)



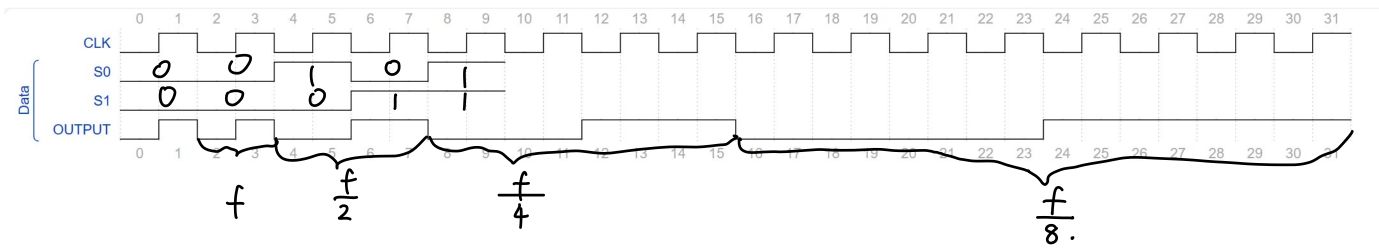

b).



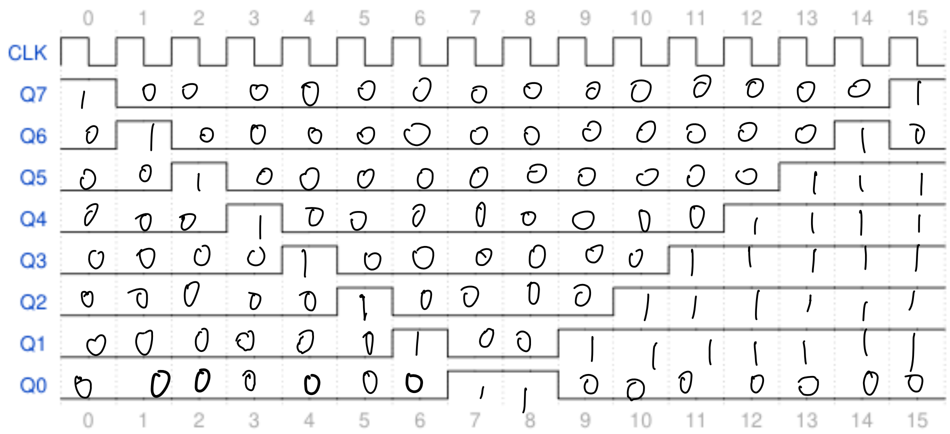
c).

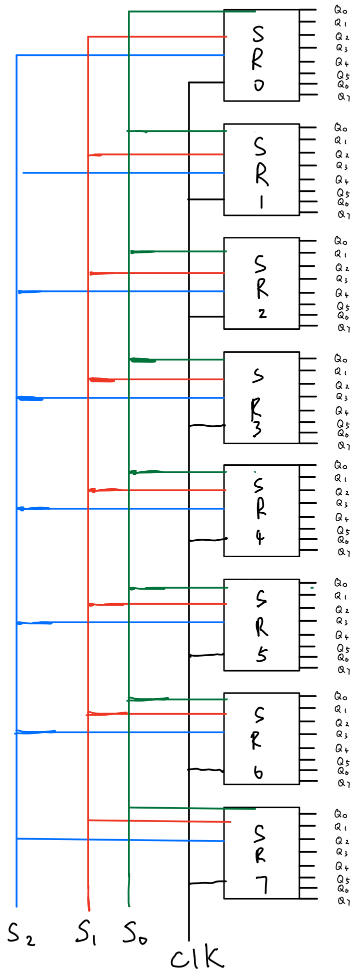

2-4-8-16
Divider
4 to 1
MUX
CLK
output.
$S_1$
$S_0$

d)


CLK
S0
S1
OUTPUT
Data
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
f
$\frac{f}{2}$
$\frac{f}{4}$
$\frac{f}{8}$.

1. Consider the timing diagram in Fig. 4. Using any combination of *shift registers* and supporting gates, create a circuit with 8-outputs that produces this timing diagram.




All SR initially set to 00000000